

$$= \frac{2.09}{100} + 2 \times \frac{0.02}{0.54} + \frac{0.002}{0.754} = 0.0976$$

$$\Delta\rho = 0.0976 \times \rho$$

$$= 0.0976 \times 5.196 \times 10^{-7}$$

$$\Delta\rho = 0.5 \times 10^{-7} \Omega \text{ m}$$

$$\rho = (5.2 \pm 0.5) \times 10^{-7} \Omega \text{ m}$$

Q1(c)

$$R = \frac{P}{I^2}$$

$$\Omega = \frac{\text{kg m}^2 \text{ s}^{-3}}{\text{A}^2} = \text{kg m}^2 \text{ s}^{-3} \text{ A}^{-2}$$

$$\therefore \Omega \text{ m} = \text{kg m}^3 \text{ s}^{-3} \text{ A}^{-2}$$

Q1(d)

The ammeter (or voltmeter) used can be changed to one of better precision, for example $\pm 0.001 \text{ A}$ instead of $\pm 0.01 \text{ A}$. Alternatively, use a cell with a greater emf.

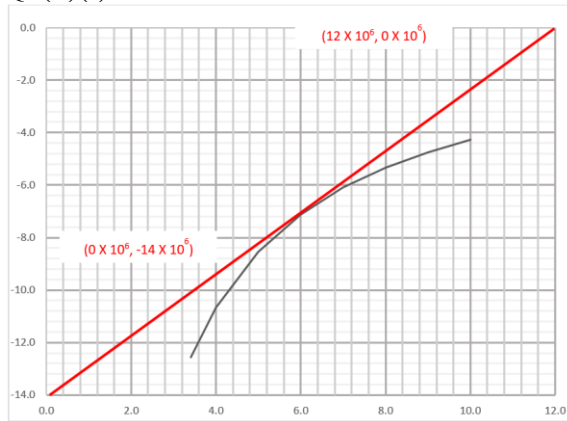
Q2(a)

Gravitational potential at infinity is defined to be zero.

The work done by an external agent to bring a mass from infinity to a point in the gravitational field is negative since the force exerted by the agent is opposite the direction of displacement from infinity.

Hence the potential at each point must be negative.

Q2(b)(i)



Gravitational Field Strength

Q2(b)(ii)

Using read offs from extrapolation

$$\begin{aligned} \text{Grad} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{(0 - (-14.0)) \times 10^6}{(12.0 - 0) \times 10^6} \end{aligned}$$

$$= 1.17 \text{ N kg}^{-1}$$

Q2(c)

$$(E_p + E_k)_{\text{initial}} = (E_p + E_k)_{\text{final}}$$

$$m\phi_{6000} + 0 = m\phi_{3390} + \frac{1}{2}mv^2$$

$$v^2 = 2(\phi_{6000} - \phi_{3390})$$

$$= 2(-7.2 - (-12.6)) \times 10^6$$

$$= 10.8 \times 10^6$$

$$v \approx 3.29 \times 10^3 \text{ m s}^{-1}$$

Q2(d)

Any of the following:

- Deploy parachutes to slow down
- Use a reverse jet to slow down
- Attach inflatable structures to help spacecraft bounce on the surface

Q2(c)(i)